

Reinforcement Learning

n -step Bootstrapping Sutton/Barto* Chapter 7

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With figures from Sutton/Barto*

*Sutton and Barto, Reinforcement Learning: An Introduction,
2nd edition, MIT Press, Cambridge, MA, 2018



Topics of this Course

- Introduction to reinforcement learning
- Markov decision processes
- **Part I: Tabular Methods**
 - Dynamic programming
 - Monte Carlo methods
 - Temporal-difference learning
 - **Multi-step bootstrapping**
 - Planning and learning with tabular methods
- Part II: Approximate Solution Methods
 - Prediction and Control using Approximation
 - Eligibility Traces
 - Policy Gradient Methods
- Part III: Modern RL Methods
 - Deep Reinforcement Learning
 - Current Applications

Summary of Notation

General

X	capital letters: random variables
x, p	lower-case letters: realizations of random variables or scalar functions
$\mathbf{w}$	Bold lower-case letters: real-valued vectors (even if random variables)
$\mathbf{W}$	bold capitals: matrices
α	Greek letters: parameters (vectors if in bold)
$\Pr\{X = x\}$	probability that a random variable X takes on the value x
$X \sim p$	random variable X selected from distribution $p(x) = \Pr\{X = x\}$
$\mathbb{E}[X]$	expectation of a random variable X , i.e., $\mathbb{E}[X] = \sum_x p(x)x$
$\operatorname{argmax}_a f(a)$	a value of action a at which $f(a)$ takes its maximal value

Value Function

G_t	return (cumulative reward) following time t
$G_{t:h}$	return from t to h (discounted and corrected)
$v_\pi(s)$	value of state s under policy π (expected return)
$v_*(s)$	value of state s under the optimal policy
$q_\pi(s, a)$	value of taking action a in state s under policy π
$q_*(s, a)$	value of taking action a in state s under the optimal policy
V, V_t	array estimates of state-value function v_π or v_*
Q, Q_t	array estimates of action-value function q_π or q_*

MDP

s, s'	states
a	an action
r	a reward
$\mathcal{S}$	set of all (nonterminal) states, $\mathcal{S}^+$ are all states
$\mathcal{A}(s)$	set of all actions available in state s
γ	discount-rate parameter
t	discrete time step
T	final time step of an episode (a.k.a. horizon)
A_t	random variable for the action at time t
S_t	random variable for the state at time t
R_t	random variable for the reward at time t
$p(s', r s, a)$	probability of transition to state s' and receiving reward r , from state s taking action a .
$p(s' s, a)$	probability of transition to state s' from state s taking action a .
$r(s, a)$	expected immediate reward from state s after action a .
$r(s, a, s')$	expected immediate reward from state s to s' with action a .
$\pi(a s)$	probability of taking action a in state s under stochastic policy π
$\pi(s)$	action taken in state s under deterministic policy π

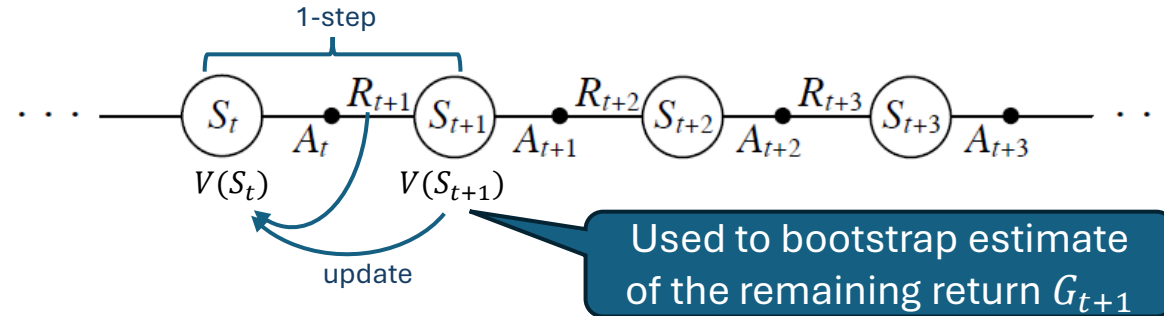
Temporal Difference Learning

U_t	target for estimate at time t
δ_t	TD error
$G_{t:t+n}$	n -step return using rewards from $t + 1$ to $t + n$ and bootstrap the rest

Policy Evaluation: 1-step TD vs. n -step TD vs. MC

Question: How far ahead in the sampled episode do we look to estimate the return?
This is a question of bias and variance of the estimate.

- **MC:** We use the whole episode. No bias, but high variance.
- **1-step TD:** We only observe the **next step** and its reward (reduces variance). The remaining return is **bootstrapped**. This **propagates state value errors (= bias)**!



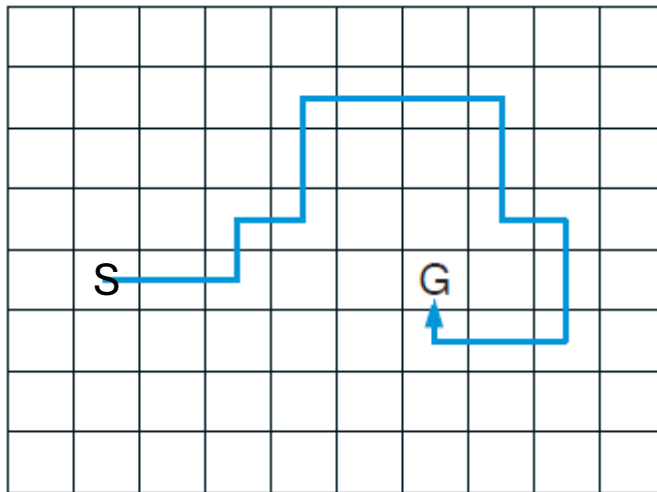
- **n -step TD**
 - Looks n time steps ahead and bootstraps the rest. Using more observed rewards should help to **reduce the bootstrapping error (bias)**, but increases variance.
 - TD unifies and generalizes MC and one-step TD.



Why is a Longer Look-ahead Useful?

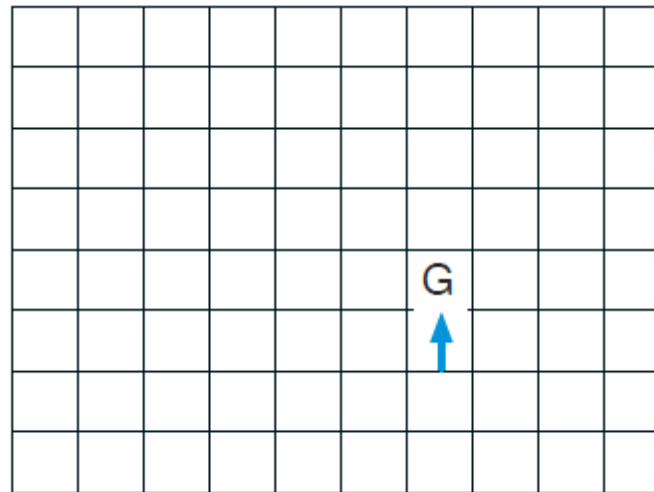
- **Example:** Maze with only one final reward for reaching the goal. We learn from the **first episode**.

Episode (MC)



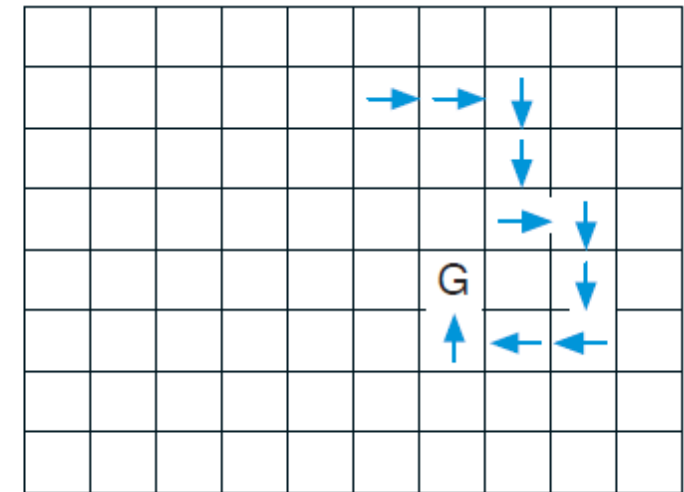
Updates in reverse order all state values on the path using the goal reward.

1-step look ahead



Can only update one state value with the goal reward.

10-step look ahead

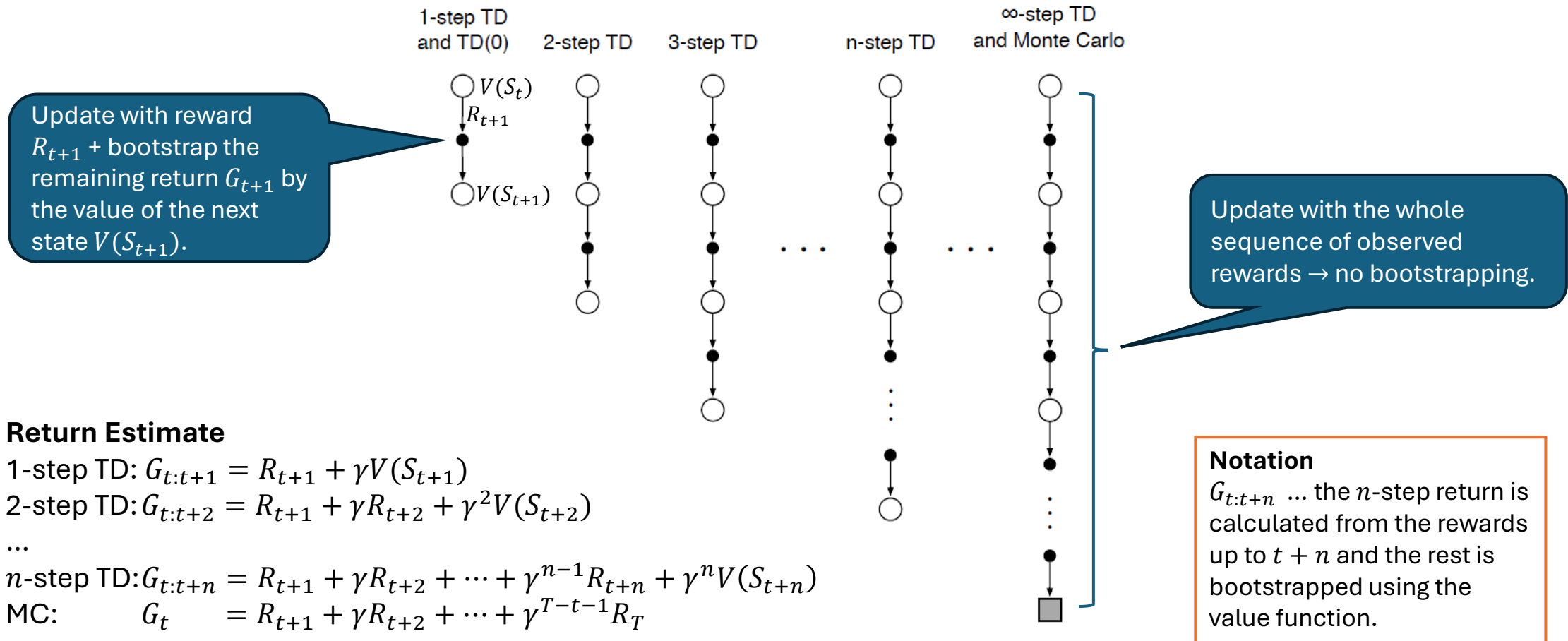


State value can be updated with the goal reward even if it is up to 10 steps away.
Learns faster than 1-step TD.

n -step TD Prediction

Estimate the value function for a given policy.

n -step TD Prediction



Return Estimate

1-step TD: $G_{t:t+1} = R_{t+1} + \gamma V(S_{t+1})$

2-step TD: $G_{t:t+2} = R_{t+1} + \gamma R_{t+2} + \gamma^2 V(S_{t+2})$

...

n -step TD: $G_{t:t+n} = R_{t+1} + \gamma R_{t+2} + \dots + \gamma^{n-1} R_{t+n} + \gamma^n V(S_{t+n})$

MC: $G_t = R_{t+1} + \gamma R_{t+2} + \dots + \gamma^{T-t-1} R_T$

Update: $V(S_t) \leftarrow V(S_t) + \alpha [G_{t:t+n} - V(S_t)]$

The update needs $G_{t:t+n}$ and is n steps delayed!

Error sources

Variance: Observed rewards

Bias: Value function estimates

Alg. n -step TD Prediction

n -step TD for estimating $V \approx v_\pi$

Input: a policy π

Algorithm parameters: step size $\alpha \in (0, 1]$, a positive integer n

Initialize $V(s)$ arbitrarily, for all $s \in \mathcal{S}$

All store and access operations (for S_t and R_t) can take their index mod $n + 1$

Loop for each episode:

Initialize and store $S_0 \neq \text{terminal}$

$T \leftarrow \infty$

Loop for $t = 0, 1, 2, \dots$:

 If $t < T$, then:

 Take an action according to $\pi(\cdot|S_t)$

 Observe and store the next reward as R_{t+1} and the next state as S_{t+1}

 If S_{t+1} is terminal, then $T \leftarrow t + 1$

$\tau \leftarrow t - n + 1$ (τ is the time whose state's estimate is being updated)

 If $\tau \geq 0$:

$G \leftarrow \sum_{i=\tau+1}^{\min(\tau+n, T)} \gamma^{i-\tau-1} R_i$

 If $\tau + n < T$, then: $G \leftarrow G + \gamma^n V(S_{\tau+n})$

$V(S_\tau) \leftarrow V(S_\tau) + \alpha [G - V(S_\tau)]$

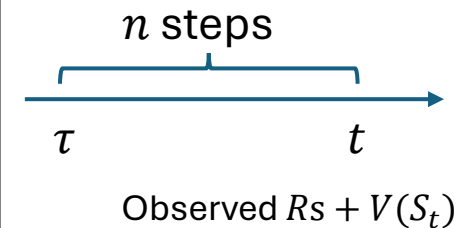
 Until $\tau = T - 1$

Detect the end of the episode

Start only when we can update $t \geq n$

The update is delayed n steps

Use bootstrapping if the episode is not over.
Note: $V(S_{\tau+n})$ is called $V(S_{t+n})$ in the equations!



Convergence Guarantee

Error Reduction Property of n -step Returns

The worst error of the expected n -step return is guaranteed to be less than or equal to γ^n times the worst error of using V .

$$\max_s |\mathbb{E}_\pi[G_{t:t+n} | S_t = s] - v_\pi(s)| \leq \gamma^n \max_s |V(s) - v_\pi(s)|$$

Explanation: $G_{t:t+n} = R_{t+1} + \gamma R_{t+2} + \dots + \gamma^{n-1} R_{t+n} + \gamma^n V(S_{t+n})$

- Observed rewards have in expectation no error.
- The error-prone bootstrap estimate is smaller because it is n time steps away and thus discounted.
- Increasing n adds more observed rewards decreases the impact of the bootstrapping error.

Note: This also means that every update is guaranteed to reduce the expected error and moves the estimate closer to the true value function.

All n -step methods (including 1-step and MC) converge!

Bias-Variance Tradeoff in RL

- We use a set of return estimates/samples G starting at state s following π as the training set $D = \{(s_1, G_1), \dots, (s_n, G_n)\}$ to learn a value function $\hat{v}_\pi(s)$ using the tabular updating rule $V(s) \leftarrow V(s) + \alpha(G_t - V(s))$.

$\hat{v}_\pi(s)$ is biased if G s are biased!

- Bias:** How far is the average prediction from the truth?

$$\text{Bias}_D[\hat{v}_\pi(s)] = \mathbb{E}_D[\hat{v}_\pi(s) - v_\pi(s)] = \mathbb{E}_D[\hat{v}_\pi(s)] - v_\pi(s)$$

- Variance:** How much do predictions fluctuate?

$\hat{v}_\pi(s)$ has initially high variance if G s have a high variance due to sampling.

$$\text{Var}_D[\hat{v}_\pi(s)] = \mathbb{E}_D \left[\left(\mathbb{E}_D[\hat{v}_\pi(s)] - \hat{v}_\pi(s) \right)^2 \right]$$

- Total error (means-squared error):** Total Error = Bias² + Var + Irreducible Noise

Method	Bias of Estimate	Variance of Estimate
MC	No bias: G_t is an unbiased sample	High: difference between sample returns; falls with sample size.
n -step	Increased bias	Reduced
TD	High bias: G_t is bootstrapped; falls when getting close to the optimal value function.	Low: only one reward is sampled

With limited data: Total error becomes min. for some n .

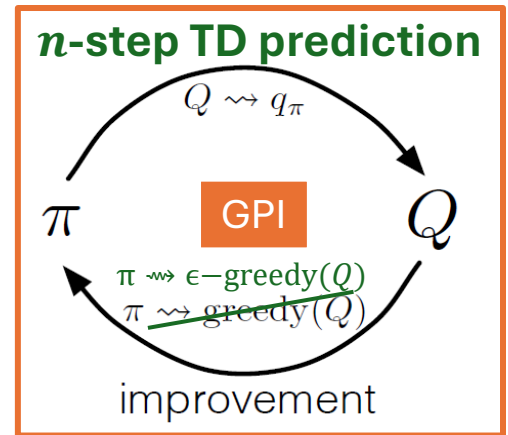
n -step Control: Sarsa

On-Policy learning using n -step return estimates.

n -step Sarsa

Note: (0) is not the number of steps but means that $\lambda = 0$. We will learn about eligibility traces later.

Original Sarsa is called one-step Sarsa or Sarsa(0)



n -step Sarsa

- **On policy:** The policy needs to keep exploring (e.g., an ϵ -greedy policy)
- Should probably be called Sarrrrr...rsa with n rs.

- n -step return estimate used as the target (redefined using Q):

$$G_{t:t+n} = R_{t+1} + \gamma R_{t+2} + \dots + \gamma^{n-1} R_{t+n} + \gamma^n Q(S_{t+n}, A_{t+n})$$

This is the extra action Sarsa need.

- Update rule:

$$Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \alpha [G_{t:t+n} - Q(S_t, A_t)]$$

Speedup of n -step Sarsa

Q -values updated during one episode

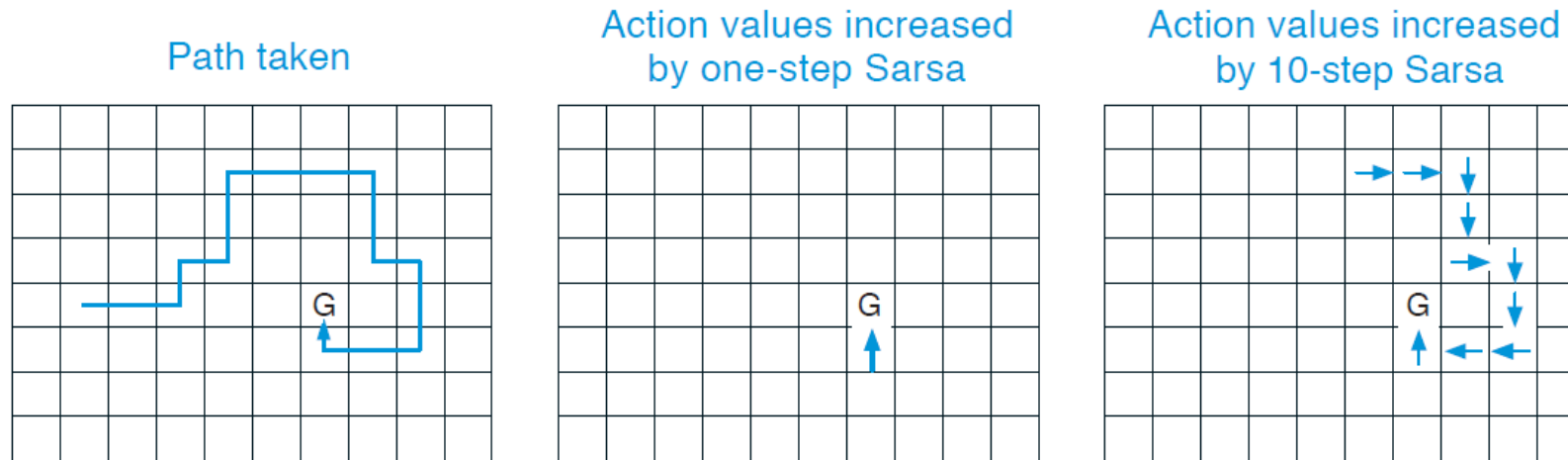


Figure 7.4: Gridworld example of the speedup of policy learning due to the use of n -step methods. The first panel shows the path taken by an agent in a single episode, ending at a location of high reward, marked by the G. In this example the values were all initially 0, and all rewards were zero except for a positive reward at G. The arrows in the other two panels show which action values were strengthened as a result of this path by one-step and n -step Sarsa methods. The one-step method strengthens only the last action of the sequence of actions that led to the high reward, whereas the n -step method strengthens the last n actions of the sequence, so that much more is learned from the one episode.

Alg. From DT prediction to n -step Sarsa Control

π is given

n -step TD for estimating $V \approx v_\pi$

Input: a policy π
 Algorithm parameters: step size $\alpha \in (0, 1]$, a positive integer n
 Initialize $V(s)$ arbitrarily, for all $s \in \mathcal{S}$
 All store and access operations (for S_t and R_t) can take their index mod $n + 1$

Loop for each episode:
 Initialize and store $S_0 \neq \text{terminal}$
 $T \leftarrow \infty$
 Loop for $t = 0, 1, 2, \dots$:
 | If $t < T$, then:
 | | Take an action according to $\pi(\cdot|S_t)$
 | | Observe and store the next reward as R_{t+1} and the next state as S_{t+1}
 | | If S_{t+1} is terminal, then $T \leftarrow t + 1$
 | $\tau \leftarrow t - n + 1$ (τ is the time whose state's estimate is being updated)
 | If $\tau \geq 0$:
 | | $G \leftarrow \sum_{i=\tau+1}^{\min(\tau+n, T)} \gamma^{i-\tau-1} R_i$
 | | If $\tau + n < T$, then: $G \leftarrow G + \gamma^n V(S_{\tau+n})$ ($G_{\tau:\tau+n}$)
 | | $V(S_\tau) \leftarrow V(S_\tau) + \alpha [G - V(S_\tau)]$
 | Until $\tau = T - 1$

Update V

n -step Sarsa for estimating $Q \approx q_\pi$ or q_*

Initialize $Q(s, a)$ arbitrarily, for all $s \in \mathcal{S}, a \in \mathcal{A}$
 Initialize π to be ε -greedy with respect to Q or to a fixed given policy
 Algorithm parameters: step size $\alpha \in (0, 1]$, small $\varepsilon > 0$, a positive integer n
 All store and access operations (for S_t, A_t , and R_t) can take their index mod $n + 1$

Loop for each episode:
 Initialize and store $S_0 \neq \text{terminal}$
 Select and store an action $A_0 \sim \pi(\cdot|S_0)$
 $T \leftarrow \infty$
 Loop for $t = 0, 1, 2, \dots$:
 | If $t < T$, then:
 | | Take action A_t
 | | Observe and store the next reward as R_{t+1} and the next state as S_{t+1}
 | | If S_{t+1} is terminal, then:
 | | | $T \leftarrow t + 1$
 | | else:
 | | | Select and store an action $A_{t+1} \sim \pi(\cdot|S_{t+1})$
 | $\tau \leftarrow t - n + 1$ (τ is the time whose estimate is being updated)
 | If $\tau \geq 0$:
 | | $G \leftarrow \sum_{i=\tau+1}^{\min(\tau+n, T)} \gamma^{i-\tau-1} R_i$
 | | If $\tau + n < T$, then $G \leftarrow G + \gamma^n Q(S_{\tau+n}, A_{\tau+n})$ ($G_{\tau:\tau+n}$)
 | | $Q(S_\tau, A_\tau) \leftarrow Q(S_\tau, A_\tau) + \alpha [G - Q(S_\tau, A_\tau)]$
 | | If π is being learned, then ensure that $\pi(\cdot|S_\tau)$ is ε -greedy wrt Q
 | Until $\tau = T - 1$

On-policy: Select next action from the current π

Update Q and π

n -step Off-Policy Control

Add importance sampling ratios to prediction and control.

n -step Off-Policy Control

- **Goal:** Learn a deterministic target policy π using an exploring (soft) behavioral policy b .
- **Method:** Use importance sampling ratio as a weight for updates

$$V_{t+1}(S_t) = V(S_t) + \alpha \rho_{t:t+n-1} [G_{t:t+n} - V(S_t)]$$

where the weight is

$$\rho_{t:h} = \prod_{k=t}^{\min(h, T-t)} \frac{\pi(A_k | S_k)}{b(A_k | S_k)}$$

How likely is π to choose an action compared to b ?

- Can be used in n -step TD for prediction or in n -step Sarsa for control.
- **Sample efficiency issue:** Can only learn (i.e., $\rho > 0$) from data where the sampled actions for the n -steps are compatible with π .
Updating the behavior policy $b = \epsilon$ -greedy(π) with a small ϵ helps.

Alg. Comparing On-Policy with Off-Policy

n -step Sarsa for estimating $Q \approx q_*$ or q_π

Initialize $Q(s, a)$ arbitrarily, for all $s \in \mathcal{S}, a \in \mathcal{A}$
 Initialize π to be ϵ -greedy with respect to Q , or to a fixed given policy
 Algorithm parameters: step size $\alpha \in (0, 1]$, small $\epsilon > 0$, a positive integer n
 All store and access operations (for S_t , A_t , and R_t) can take their index mod $n + 1$

Loop for each episode:
 Initialize and store $S_0 \neq \text{terminal}$
 Select and store an action $A_0 \sim \pi(\cdot | S_0)$
 $T \leftarrow \infty$
 Loop for $t = 0, 1, 2, \dots$:

On-Policy

If $t < T$, then:
 Take action A_t
 Observe and store the next reward as R_{t+1} and the next state as S_{t+1}
 If S_{t+1} is terminal, then:
 $T \leftarrow t + 1$
 else:
 Select and store an action $A_{t+1} \sim \pi(\cdot | S_{t+1})$

$\tau \leftarrow t - n + 1$ (τ is the time whose estimate is being updated)
 If $\tau \geq 0$:
 $G \leftarrow \sum_{i=\tau+1}^{\min(\tau+n, T)} \gamma^{i-\tau-1} R_i$
 If $\tau + n < T$, then $G \leftarrow G + \gamma^n Q(S_{\tau+n}, A_{\tau+n})$ ($G_{\tau:\tau+n}$)
 $Q(S_\tau, A_\tau) \leftarrow Q(S_\tau, A_\tau) + \alpha [G - Q(S_\tau, A_\tau)]$
 If π is being learned, then ensure that $\pi(\cdot | S_\tau)$ is ϵ -greedy wrt Q

Until $\tau = T - 1$

Off-policy n -step Sarsa for estimating $Q \approx q_*$ or q_π

Input: an arbitrary behavior policy b such that $b(a|s) > 0$, for all $s \in \mathcal{S}, a \in \mathcal{A}$
 Initialize $Q(s, a)$ arbitrarily, for all $s \in \mathcal{S}, a \in \mathcal{A}$
 Initialize π to be greedy with respect to Q , or as a fixed given policy
 Algorithm parameters: step size $\alpha \in (0, 1]$, a positive integer n
 All store and access operations (for S_t , A_t , and R_t) can take their index mod $n + 1$

Loop for each episode:
 Initialize and store $S_0 \neq \text{terminal}$
 Select and store an action $A_0 \sim b(\cdot | S_0)$
 $T \leftarrow \infty$
 Loop for $t = 0, 1, 2, \dots$:

If $t < T$, then:
 Take action A_t
 Observe and store the next reward as R_{t+1} and the next state as S_{t+1}
 If S_{t+1} is terminal, then:
 $T \leftarrow t + 1$
 else:
 Select and store an action $A_{t+1} \sim b(\cdot | S_{t+1})$

$\tau \leftarrow t - n + 1$ (τ is the time whose estimate is being updated)
 If $\tau \geq 0$:
 $\rho \leftarrow \prod_{i=\tau+1}^{\min(\tau+n, T-1)} \frac{\pi(A_i | S_i)}{b(A_i | S_i)}$ ($\rho_{\tau+1:\tau+n}$)
 $G \leftarrow \sum_{i=\tau+1}^{\min(\tau+n, T)} \gamma^{i-\tau-1} R_i$ ($G_{\tau:\tau+n}$)
 If $\tau + n < T$, then: $G \leftarrow G + \gamma^n Q(S_{\tau+n}, A_{\tau+n})$ ($G_{\tau:\tau+n}$)
 $Q(S_\tau, A_\tau) \leftarrow Q(S_\tau, A_\tau) + \alpha \rho [G - Q(S_\tau, A_\tau)]$
 If π is being learned, then ensure that $\pi(\cdot | S_\tau)$ is greedy wrt Q

Until $\tau = T - 1$

Add behavior policy

Greedy target policy

We could update $b = \epsilon$ -greedy(π)

Use importance sampling ratio



What you Need to Know

- n -step methods let us choose the amount of bootstrapping from 1-step to no bootstrapping (MC).
- Advantage over one-step methods:
 - **Learning:** Looking more steps ahead speeds up learning, especially for sparse rewards.
 - **Error reduction:** Significantly reduces the bootstrapping error (**bias**) at the cost of increased **variance**.
 - An intermediate amount of bootstrapping typically **performs better** in terms of the variance/bias tradeoff than the one-step and MC extremes. n is application-dependent and can be tuned.
- Drawback:
 - **Delay:** Updates are delayed n -step since future events (rewards and states) are needed.
 - **Computation:** More computation per time step and higher memory needs than one-step methods.

Eligibility traces address the delay and computation drawbacks. We will cover these methods later.